

A LYAPUNOV THEOREM FOR THE DE BRUIJN–NEWMAN FLOW, A DYNAMICAL NO-GO, AND THE EXTREMAL FORM OF THE RIEMANN HYPOTHESIS AS GAP UNIVERSALITY

DAVID ALEJANDRO TREJO PIZZO

ABSTRACT. We study the Riemann Hypothesis through the de Bruijn–Newman heat flow, for which $\text{RH} \iff \Lambda = 0$ (using Rodgers–Tao’s $\Lambda \geq 0$ and Polymath15’s $\Lambda \leq 0.2$). We record three RH-independent results. (i) A clean *Lyapunov theorem*: along the Coulomb-gas zero dynamics, the total off-line-ness $\mathcal{L} = \sum_j (\text{Im } z_j)^2$ obeys $\dot{\mathcal{L}} = -2 \sum_{j \neq k} (y_j - y_k)^2 / |z_j - z_k|^2 \leq 0$, so $\mathcal{L}$ is a strict Lyapunov function and the critical line is the flow’s global attractor; as a by-product any hypothetical off-line zero sits at imaginary height $\leq \sqrt{2\Lambda} \leq 0.63$. (ii) A *dynamical no-go* (N5): the identity for $\dot{\mathcal{L}}$ is *arithmetic-blind* (it holds for every conjugate-symmetric configuration), so no argument using the flow alone can prove RH; in particular this subsumes the entire class of “critical-line-as-attractor” dissipative-dynamics / PT-symmetry heuristics. (iii) The *extremal reduction*: Λ equals the first backward-collision time of the zeros under the flow, an extremal (not average) quantity, and a single off-line zero already forces $\Lambda > 0$. Measuring real ζ zeros we find the normalized backward-collision margin t_c/s^2 clusters at a universal negative constant ≈ -0.02 (the sine-kernel prediction), giving the sharpest statement of the target: *RH follows from the unconditional GUE gap universality of the zeros*. We locate, precisely and quantitatively, the single irreducible arithmetic statement at which the dynamical approach—and, as a companion map shows, every current analytic approach—arrives.

CONTENTS

1.	Introduction	2
2.	A Lyapunov theorem and the attractor	2
3.	A dynamical no-go: the flow is arithmetic-blind	3
4.	The extremal reduction, and the target as gap universality	3
5.	Conclusion	4
	References	4

1. INTRODUCTION

The de Bruijn–Newman constant Λ is defined through the one-parameter deformation

$$(1) \quad H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du, \quad H_0(z) \propto \xi\left(\frac{1}{2} + \frac{iz}{2}\right),$$

where ξ is the Riemann ξ -function and Φ its (super-exponentially decaying) kernel; H_t satisfies the backward heat equation $\partial_t H_t = -\partial_{zz} H_t$. By the theorems of de Bruijn and Newman, there is a constant $\Lambda \in \mathbb{R}$ such that H_t has only real zeros iff $t \geq \Lambda$. Newman conjectured $\Lambda \geq 0$ (“RH, if true, is only barely so”); this was proved by Rodgers and Tao [1], while Polymath15 [2] established $\Lambda \leq 0.2$. Consequently

$$(2) \quad \text{RH} \iff \Lambda \leq 0 \iff \Lambda = 0.$$

This reframes RH as the statement that the flow’s zeros are *already* all real at $t = 0$, the boundary of the all-real regime. It is attractive because the reality of the zeros for $t \geq \Lambda > 0$ is a *theorem*, not RH, and the only remaining content is the propagation of reality down to $t = 0$, governed by the zero dynamics rather than by the explicit formula.

This paper records what that dynamical viewpoint yields rigorously. The results are RH-independent: a Lyapunov theorem and its attractor corollary (§2), a no-go showing the flow alone cannot decide RH (§3), and the extremal reduction of Λ to a backward-collision time together with the numerical observation that pins the target as gap universality (§4). The companion paper [5] places these alongside the static reformulations; here we develop the dynamical side in full.

The zero dynamics. Where H_t has simple zeros $z_j(t)$, differentiating $H_t(z_j(t)) = 0$ and using $\partial_t H = -\partial_{zz} H$ gives the Calogero/Coulomb flow

$$(3) \quad \dot{z}_j = 2 \sum_{k \neq j} \frac{1}{z_j - z_k},$$

the classical zero dynamics of the heat flow [3]. (We normalize the time so the constant is 2; all statements below are invariant under rescaling t .) Since H_t has real coefficients, the zeros are symmetric under conjugation; complex zeros occur in conjugate pairs.

2. A LYAPUNOV THEOREM AND THE ATTRACTOR

Write $z_j = x_j + iy_j$ and define the *total off-line-ness*

$$(4) \quad \mathcal{L} := \sum_j (\text{Im } z_j)^2 = \sum_j y_j^2 \geq 0, \quad \mathcal{L} = 0 \iff \text{all zeros real.}$$

Theorem 2.1 (Lyapunov monotonicity). *Along the flow (3),*

$$(5) \quad \dot{\mathcal{L}} = -2 \sum_{j \neq k} \frac{(y_j - y_k)^2}{|z_j - z_k|^2} \leq 0,$$

with equality iff all y_j are equal; for a conjugate-symmetric configuration this forces all $y_j = 0$. Hence $\mathcal{L}$ is a strict Lyapunov function for the forward flow, and the set of all-real configurations is its global attractor.

Proof. $\dot{\mathcal{L}} = 2 \sum_j y_j \dot{y}_j = 2 \sum_j y_j \text{Im } \dot{z}_j$. By (3), $\text{Im } \dot{z}_j = 2 \sum_{k \neq j} \text{Im}(z_j - z_k)^{-1} = -2 \sum_{k \neq j} (y_j - y_k) / |z_j - z_k|^2$, so $\dot{\mathcal{L}} = -4 \sum_{j \neq k} y_j (y_j - y_k) / |z_j - z_k|^2$. Symmetrizing under $j \leftrightarrow k$ (each unordered pair twice) replaces $y_j (y_j - y_k)$ by $\frac{1}{2} [(y_j - y_k)^2]$, giving (5). Each summand is ≥ 0 ; equality needs $y_j = y_k$ for all j, k , and conjugate symmetry ($\sum y_j = 0$, y ’s in $\pm$ pairs) then forces $y_j \equiv 0$. $\square$

This is the rigorous content of the intuition that “the critical line is a stable attractor”: under the heat flow, off-line zeros are pulled toward the real axis, monotonically in the quadratic measure $\mathcal{L}$. We record two consequences.

Corollary 2.2 (Healing rate; height bound). *An isolated conjugate pair at $\pm iy_0$ obeys $\frac{d}{dt}y_0^2 = -2$ exactly, hence heals (reaches the real axis) by time $y_0^2/2$. Therefore any zero of $H_0 = \xi$ off the real axis (i.e. any RH-violating zero) has imaginary part $|y_0| \leq \sqrt{2\Lambda} \leq \sqrt{0.4} \approx 0.63$.*

Proof. For the isolated pair, (5) gives $\dot{\mathcal{L}} = -2 \cdot 2 \cdot \frac{(2y_0)^2}{(2y_0)^2} = -4$ and $\mathcal{L} = 2y_0^2$, so $\frac{d}{dt}y_0^2 = -2$ and $y_0^2(t) = y_0^2(0) - 2t$. A pair present at $t = 0$ heals by $t = y_0^2(0)/2$; since the all-real regime begins at $t = \Lambda$, $y_0^2(0)/2 \leq \Lambda$, and $\Lambda \leq 0.2$. $\square$

3. A DYNAMICAL NO-GO: THE FLOW IS ARITHMETIC-BLIND

Theorem 2.1 is appealing, but it cannot prove RH, for two structural reasons that we make precise.

First, *wrong direction*: forward in t , $\mathcal{L}$ decreases to 0 (the regime $t \geq \Lambda$). RH concerns $t = 0 \leq \Lambda$, reached by flowing *backward*, where $\mathcal{L}$ *increases*; the monotonicity then yields only $\mathcal{L}(0) \geq \mathcal{L}(\Lambda) = 0$, which is trivially true and not $\mathcal{L}(0) = 0$.

Second, and decisively, *arithmetic-blindness*.

Theorem 3.1 (Dynamical no-go, N5). *The identity (5) holds for every finite conjugate-symmetric configuration $\{z_j\}$, independently of any arithmetic origin. Consequently no argument that uses only the flow (3) together with a Lyapunov/monotonicity functional of the zero positions can distinguish ξ from an arbitrary such function, and hence cannot prove RH: it would prove “every function in the de Bruijn–Newman class satisfies RH,” which is false (generic initial data have $\Lambda > 0$). Any RH attack based on a dissipative dynamics with the critical line as attractor is of this form, and is subject to this obstruction.*

Proof. The derivation of (5) used only (3) and $z_j = x_j + iy_j$, never the values $\{z_j(0)\}$. The flow (3) is the same vector field for all initial data. A functional $F(\{z_j\})$ of the positions therefore has $\dot{F}$ determined by the universal field; only the initial value $F(\{z_j(0)\})$ carries the arithmetic, and that value is exactly what RH asserts. There exist functions in the class with $\Lambda > 0$ (off-line zeros), e.g. suitable Fourier transforms violating the Hermite–Biehler condition; for these the same $\dot{\mathcal{L}} \leq 0$ holds. Hence the flow cannot separate $\Lambda \leq 0$ from $\Lambda > 0$. $\square$

Remark 3.2. Theorem 3.1 explains, in a single sentence, why the heat flow does not escape the difficulty even though it introduces a genuinely new parameter t : the new degree of freedom is arithmetic-blind. RH lives entirely in the *initial condition* $H_0 = \xi$, i.e. in where the zeros start and how they are spaced. To use the flow one must re-inject ζ ’s zero statistics; §4 identifies exactly which statistic.

4. THE EXTREMAL REDUCTION, AND THE TARGET AS GAP UNIVERSALITY

Because the flow preserves the real axis forward but contracts gaps backward, Λ admits a clean extremal description.

Proposition 4.1 (Λ as first backward collision). *Under (3) a pair of real zeros with gap g obeys $g(t)^2 = g_0^2 + 8t$ to leading (two-body) order, so backward in t the gap shrinks and the pair collides (forming a double zero, then a conjugate pair) at $t_c = -g_0^2/8$. The de Bruijn–Newman constant is the supremum of such collision times over all pairs and heights: $\Lambda = \sup t_c$. In particular a single off-line zero forces $\Lambda > 0$; Λ is an extremal, not an averaged, functional of the zeros.*

This is the crux. Any control of RH through the flow must bound the *worst* pair, uniformly in height — no averaged statistic (mean spacing, integrated pair correlation) suffices, since each is insensitive to a sparse set of anomalously tight pairs, while one such pair decides Λ .

A universal collision margin (numerical). We backward-flow blocks of consecutive real ζ zeros (via `mpmath.zetazero`) until the tightest pair collides, recording the collision time t_c and the local mean spacing $s \sim 2\pi/\log T$:

height T	min gap	mean spacing s	t_c	t_c/s^2
99	0.845	2.281	−0.105	−0.0202
850	0.484	1.280	−0.032	−0.0195
2547	0.425	1.046	−0.025	−0.0229

In absolute units the margin $|t_c|$ shrinks with height (the spacing $s \rightarrow 0$), but in mean-spacing units t_c/s^2 clusters at a *universal negative constant* ≈ -0.02 . This is the sine-kernel / GUE prediction for the (normalized) closest-pair collision time, and it is consistent with the marginal value $\Lambda = 0$: the supremum of t_c tends to 0^- in absolute units while the normalized margin stays strictly below 0. (The computation is a deliberate probe—three blocks, a windowed flow without the mean field—so the constant is suggestive, not established.)

Observation 4.2 (The sharpest target). By Proposition 4.1, $\text{RH} = \{\Lambda \leq 0\}$ follows if the normalized backward-collision margin t_c/s^2 stays a negative constant uniformly in T — equivalently, if the closest-pair gap statistics of the ζ zeros are GUE-universal *unconditionally*. This is the extremal, quantitative form of the target. Its uniform control is precisely the open, currently RH-conditional, statement of pair-correlation / gap universality (Montgomery’s conjecture is proved only in a restricted range [4]).

5. CONCLUSION

The de Bruijn–Newman flow gives a clean dynamical picture of RH and three RH-independent results: a Lyapunov theorem with the critical line as attractor (Theorem 2.1) and the height bound (Corollary 2.2); a dynamical no-go showing the flow alone is arithmetic-blind and subsuming the PT-symmetry/dissipative heuristics (Theorem 3.1); and the extremal reduction of Λ to a backward-collision time, with the numerical universality of the normalized margin pinning the target as the *unconditional gap universality* of the zeros (Proposition 4.1, Observation 4.2). The viewpoint does not cross the wall; it compresses RH, with unusual precision, into a single extremal arithmetic statement. Establishing that statement unconditionally—the GUE universality of the closest-pair gaps, uniform in height—is the new analytic input the problem demands, and the subject of a separate program.

REFERENCES

- [1] B. Rodgers and T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum Math. Pi **8** (2020), e6.
- [2] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, Res. Math. Sci. **6** (2019), 31.
- [3] G. Csordas, W. Smith, R. S. Varga, *Lehmer pairs of zeros, the de Bruijn–Newman constant Λ , and the Riemann hypothesis*, Constr. Approx. **10** (1994), 107–129.
- [4] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. **24** (1973), 181–193.
- [5] D. A. Trejo Pizzo, *The band-limited Weil–Carleson form and a five-language map of the Riemann wall*, preprint, 2026.
- [6] C. M. Newman, *Fourier transforms with only real zeros*, Proc. Amer. Math. Soc. **61** (1976), 245–251.
- [7] G. Csordas, W. Smith, and R. S. Varga, *Lehmer pairs of zeros, the de Bruijn–Newman constant Λ , and the Riemann Hypothesis*, Constr. Approx. **10** (1994), 107–129.